



Non-contact blood pressure estimation using FMCW radar: A two-stream approach focused on central arterial activity

Zhongrui Bai^{a,b}, Fanglin Geng^b, Hao Zhang^b, Xianxiang Chen^b, Lidong Du^b, Peng Wang^b, Pang Wu^{b,*}, Gang Cheng^{c,*}, Zhen Fang^{b,d,**}, Yirong Wu^{b,*}

^a School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, 200240, China

^b Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, 100190, China

^c Department of Neurosurgery, First Medical Center of Chinese PLA General Hospital, Beijing, 100080, China

^d Personalized Management of Chronic Respiratory Disease, Chinese Academy of Medical Sciences, Beijing, 100190, China

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ABSTRACT

This paper proposes a radar-based two-stream blood pressure (BP) estimation framework (R2S-BP), focusing on central arterial activity. It separately analyzes central-arterial pulse transit time (caPTT) and pulse wave morphology using multi-location Doppler Cardiogram (DCG) data from millimeter wave FMCW radar. Specifically, phase information at harmonic heart rate frequencies is used to compute time delay arrays, representing caPTT-related features. Additionally, k-Shape clustering is employed to select optimal DCGs from the neck and chest regions that contain BP-related morphological features. These features are processed through a two-stream neural network combining BiLSTM, ResNet, and multi-head attention modules. Subject-independent 9-fold cross-validation results show that the standard deviations of the errors for systolic and diastolic BP are 7.33 and 5.36 mmHg, respectively. The intra-subject correlation coefficient for both systolic and diastolic BP averages 0.82. Comparative and ablation studies demonstrate the superiority of the two-stream approach and the critical importance of its components. This approach integrates physiologically guided manual feature construction with a deep learning model, fully leveraging the capabilities of FMCW radar data.

1. Introduction

Blood pressure (BP) is a critical indicator of overall health, particularly in relation to the cardiovascular system. Recent studies have demonstrated that dynamic BP, especially nighttime BP, offers more insights into all-cause mortality and cardiovascular mortality risks than snapshot BP measurements [1].

However, commonly used dynamic or continuous BP measurement methods, such as oscillometric BP monitors, arterial catheterization, and constant volume methods, are not suitable for long-term BP monitoring due to their invasiveness, discomfort, and potential to disrupt sleep. This has led researchers to explore unobtrusive, dynamic, or continuous BP monitoring methods. These new approaches can be broadly categorized into two types based on their principles: Pulse Transit Time (PTT) and Pulse Wave Analysis (PWA). PTT measures the delay between pulse waves at different locations to correlate with BP [2–4], while PWA uses the shape, amplitude, and other features of

the pulse wave to estimate BP [5,6].

Regardless of whether PTT or PWA methods are used, photoplethysmography (PPG) is the most commonly used method for obtaining pulse waves. However, when using finger PPG as the distal pulse wave for PTT calculation, the presence of smooth muscle (SM) in peripheral arteries, which is regulated by neurohumoral control mechanisms on a timescale of seconds to minutes, results in vessel characteristics that are independent of BP variations [7]. Therefore, PPG-based PTT methods require frequent calibration. Additionally, when using PPG for PWA, the complex pressure-volume relationship and influences such as SM, sensor contact pressure, and ambient light limit its reliability [7].

Fortunately, in recent years, the widespread use of radar in vital sign monitoring has provided new solutions for cuffless BP monitoring [8]. Radar obtains pulse waves through the modulation of electromagnetic waves by heart and arterial movements. Compared to

* Corresponding authors.

** Corresponding author at: Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, 100190, China.

E-mail addresses: zhongrui.bai@sjtu.edu.cn (Z. Bai), gengfanglin20@mails.ucas.ac.cn (F. Geng), zhanghao190@mails.ucas.ac.cn (H. Zhang), chenxx@aircas.ac.cn (X. Chen), lddu@mail.ie.ac.cn (L. Du), wangpeng01@aircas.ac.cn (P. Wang), wupang@aircas.ac.cn (P. Wu), yjscg2003@126.com (G. Cheng), fangzhen@aircas.ac.cn (Z. Fang), wyr@mail.ie.ac.cn (Y. Wu).

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PPG, radar acquires mechanical signals directly, avoiding the volume-to-pressure conversion during BP measurement and facilitating the capture of aortic rather than capillary information. In this paper, the heart-induced micro-vibrations detected by radar are referred to as Doppler Cardiogram (DCG) [9,10].

The connection between DCG and BP has been explored by numerous studies. Early studies primarily focused on contact-based radar BP estimation. For example, Buxi et al. [2] placed a continuous wave radar on the sternum to capture arterial pulsations from the aortic arch, using bioimpedance and ECG on the shoulder to estimate BP via PTT between different pulse waves. Liao et al. [11] used a wrist-mounted Frequency Modulated Continuous Wave (FMCW) radar to measure radial artery pulsations, converting the radar-measured pulse wave units directly into pressure units (mmHg) using initial calibration with reference BP values. Wen et al. [12] used a 120 GHz continuous wave (CW) radar fixed at the elbow to measure brachial artery pulse waves, estimating BP through the reflective pulse transit time. These contact-based measurement methods still cause users discomfort and are not suitable for daily monitoring scenarios.

Recent studies have explored non-contact radar BP estimation, such as Shi et al. [13] using FMCW radar to non-invasively measure wrist pulse waves, estimating BP through features extracted from the delayed Doppler domain and random forest regression models. Liang et al. [5] also used FMCW radar to measure radial artery pulsations, obtaining pulse waves through beamforming optimized by ACC-Max and estimating BP with multi-layer attention networks. Recently, Qiu et al. [14] applied a stacked deformable convolution network for BP estimation based on CW radar.

While these existing studies provide valuable references for non-contact radar-based BP estimation, they either use additional contact sensors for multi-site pulse wave measurement or estimate BP through single-site pulse wave analysis, which poses limitations in comfort or reliability.

In fact, FMCW radar offers distance and angle resolution, providing more information related to cardiac movements and arterial pulsations. In our previous study [15], we proposed a method for measuring central-arterial Pulse Transit Time (caPTT) using a single FMCW radar. Distinct from other radar-based methodologies that predominantly focus on peripheral wrist pulse waves, our system positioned a millimeter-wave FMCW radar with MIMO antennas directly above the neck, allowing for the simultaneous acquisition of DCG signals from both the neck and chest. This configuration allows for the calculation of caPTT, which has been shown to be more stable than peripheral PTT due to the lower SM content in the aorta compared to peripheral arteries. Through validation experiments involving the synchronous acquisition of multiple physiological signals such as electrocardiogram (ECG), phonocardiogram (PCG), PPG, and carotid pulse waves, we demonstrated a high correlation of 0.91 between radar-derived PTT and contact-based PTT measurements. Subsequently, we employed regression fitting methods to compute beat-by-beat BP, confirming the relationship between radar-obtained central arterial caPTT and BP.

However, our preliminary study has several limitations. First, the extracted signals were not limited to the intended neck or chest region but included a broader body area due to angular signal leakage from beam separation, which compromised the accuracy of caPTT extraction. Second, our preliminary study focused solely on caPTT without utilizing the morphological characteristics of the DCG. Third, selecting the desired signals from multiple DCG sources and identifying feature points required significant manual intervention due to the instability and variability of DCG morphology. This instability and variability arise from the variations in arterial waveforms along the arterial tree, as well as mutual interference between DCGs from different ranges and angles. Consequently, these limitations hinder the accuracy of BP estimation and reduce the method's overall usability.

Fortunately, deep learning offers potential solutions to address these challenges. It has demonstrated promising performance in the field

of physiological signal sensing [16], particularly in radar-based applications [17]. In various physiological signal sensing tasks, such as emotion recognition [18], heart rate variability estimation [19], sleep staging [20], and BP estimation [6,14], well-constructed deep learning models have shown remarkable performance. On the other hand, several studies have indicated that incorporating prior physiological knowledge into the construction of model input features can significantly enhance both the model's performance and robustness [21].

In this work, we aim to address the aforementioned issues and extend our previous research. To fully exploit the body vibrations captured by the radar, we propose *Radar-based Two-Stream* method for BP estimation (R2S-BP). R2S-BP first processes FMCW radar data through a dedicated processing chain, resulting in a DCG array from various body parts. Subsequently, two types of network inputs containing caPTT and central-arterial Pulse Wave Analysis (caPWA) information are designed and extracted. The Multi-Location DCG Harmonic Phase Array (MLD-HPA) and the Pulse Delay Array (PDA) are first extracted as caPTT information. The two optimal DCGs from the neck and chest regions, obtained via k-Shape clustering, serve as PWA information. These two types of information are then input into a two-stream network comprising BiLSTM, ResNet, and a multi-head attention module. Data augmentation in the range and angle dimensions is applied to enhance data diversity and model robustness. Additionally, a Model-Agnostic Meta-Learning (MAML)-based training strategy is used for calibration-based experiments with independent subjects. The superiority of the proposed method is demonstrated through multiple experiments involving data induced by three different methods.

The contributions of this paper can be summarized as follows:

- (1) This study proposes a two-stream framework for FMCW radar-based BP estimation, which meticulously focuses on and analyzes PTT and PWA-related information derived from radar-captured central artery activity data.
- (2) This study proposes a pulse delay array extraction method based on the phases of fundamental frequency and harmonic of heartbeats, as well as a k-Shape-based dimensionality reduction method for the DCG array. These methodologies effectively construct robust caPTT and caPWA information.
- (3) A two-stream neural network comprising BiLSTM, ResNet, and multi-head attention is designed to extract and integrate different types of features. Additionally, a data augmentation strategy tailored for FMCW radar physiological data is employed to enhance data diversity and model robustness.
- (4) The proposed method's superiority is validated through multiple experiments involving data from 36 subjects, induced by three different methods of BP variation, using a MAML-based subject-independent scheme.

The remainder of this paper is organized as follows: Section 2 provides a detailed description of the proposed R2S-BP method. Section 3 describes the dataset used, implementation details, and evaluation metrics. Section 4 presents and discusses the experimental results. Section 5 concludes the study.

2. Method

2.1. Overview

Fig. 1 illustrates the overall workflow of the proposed R2S-BP framework. The framework comprises the following processes:

Data acquisition. According to our predetermined experimental protocol, FMCW radar was used to collect radar echo data from 40 cm above the subjects' necks, focusing on central arterial activity. For more details, please refer to Section 3.1 and our previous work [15].

Radar data processing. The collected radar echo data underwent a MIMO-FMCW data processing pipeline. Subsequently, signals of body surface vibrations (DCGs) from different distances and different angles are obtained. Data selection and augmentation are then performed on

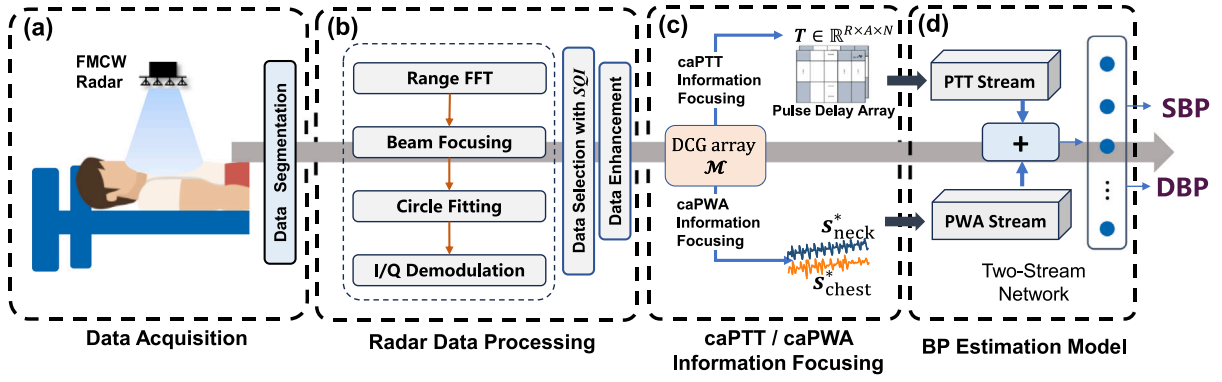


Fig. 1. Overview of R2S-BP. (a) A MIMO-FMCW radar is suspended above the neck to capture detailed motion information spanning from the neck to the chest. (b) Radar echo data undergoes processing through a customized radar signal processing chain, followed by data selection and augmentation procedures, resulting in the acquisition of a DCG array. (c) Considering the variability in DCG signal morphology, methods are designed to extract features related to caPTT and caPWA. (d) A customized dual-stream neural network is then used to analyze and integrate these features to estimate BP.

the DCGs. This process is detailed in Section 2.2.

caPTT/caPWA information focusing. PTT and PWA are two types of information effective for BP estimation, both of which are contained within the DCG array captured by FMCW radar from the human neck to the chest. To extract useful information from the high-dimensional DCG array for BP estimation, different methods were developed to derive pulse transit time-related information (referred to as caPTT information) and pulse wave morphology-related information (referred to as caPWA information). The specific details are described in Section 2.3.

BP estimation model. To map the extracted caPTT and caPWA information to BP, a dual-stream neural network model was designed. The two types of data are input separately into the dual-stream network's feature extraction branches. In the final part of the network, the features are fused to generate outputs for SBP and DBP. Additionally, a MAML-based training strategy is employed to facilitate subject-independent experiments. This process is detailed in Sections 2.4 and 2.5.

2.2. Radar data processing

2.2.1. FMCW radar processing chain

FMCW radar determines distance by analyzing the frequency difference between transmitted and received signals. It detects μm -level movement of body surface caused by heartbeat and respiration by calculating phase variations of intermediate frequency signals at specific ranges. Additionally, this work employs MIMO radar to capture data from eight virtual channels aligned along the head-foot's axis.

As illustrated in Fig. 1(b), the signal processing chain of the FMCW radar in this study includes Range FFT, Beam Focusing, Circle Fitting, and I/Q Demodulation. For more details, refer to previous work [15]. Notably, there are two key differences in this study. First, during the Beam Focusing stage, this study calculates the DCG for all positions within the region of interest, rather than focusing solely on the chest and neck. Second, during the I/Q Demodulation stage, Modified Differentiate and Cross-Multiply (MDACM) [22] is used instead of arctangent demodulation to avoid baseline drift caused by phase unwrapping, while also combating noise. In the digital domain, the MDACM algorithm is represented as follows:

$$x[n] = \frac{\lambda}{4\pi} \sum_{k=2}^n (I[k-1]Q[k] - I[k]Q[k-1]), \quad (1)$$

where n , k denote the integration length and index of the summation, respectively. $x[n]$, $I[k]$, and $Q[k]$ represent the digital forms of the target displacement, the in-phase component, and the quadrature component of the intermediate frequency (IF) signal, respectively. In this work, with a frame rate of 250 Hz, the maximum integration length is limited to 150 to filter out respiratory components. The resulting

demodulated signal is then differentiated once more to further extract the heartbeat component. Finally, according to our experimental setup, we selected data within the range from the 7th to the 11th and angles from 70 degrees to 142 degrees, denoted as $\mathcal{M}^0 \in \mathbb{R}^{5 \times 72 \times L}$, for the next processing step.

2.2.2. Data selection with SQI

To account for the influence of movements, speech, and other factors during the experimental process, it is necessary to select $\mathcal{M}^0$ with high signal quality, ensuring that only data samples containing clean heartbeat signals are retained. In this study, we designed and employed a Signal Quality Index (SQI) using a harmonic summation algorithm, similar to those used in speech fundamental frequency estimation [23], to evaluate the quasi-periodicity of the signals without requiring additional reference signals such as ECG.

First, we calculate the FFT spectrum of each DCG signal s in $\mathcal{M}^0$. Then, we compute its Harmonic Summation (HS) spectrum from the FFT spectrum. Specifically, for each candidate fundamental frequency f , the value of its HS spectrum is calculated as the inner product of the FFT spectrum with a kernel function corresponding to f :

$$HS(f) = \langle |\mathcal{F}_s(v)|, K(f, v) \rangle, \quad (2)$$

where the notation $\langle \cdot, \cdot \rangle$ signifies the inner product operation. $|\mathcal{F}_s(v)|$ is the amplitude frequency response of s from FFT, with v representing the frequency in the new coordinate. $K(f, v)$ is the kernel function, defined by

$$K(f, v) = \begin{cases} 1 & \text{if } v = n \times f \\ -1/2 & \text{if } v = (n \pm 1/2) \times f, \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

and $n \in 1, 2, \dots, 6$, as the 6 harmonics of the previous heartbeat can contain most of the energy of the heartbeat according to observations. Then, as shown in Fig. 2, we identify the frequency f_{HR} at which $HS(f)$ reaches its maximum value within the range of 0.5 Hz to 3.5 Hz, which is the possible heart rate range:

$$f_{\text{HR}} = \arg \max_{0.5 \text{ Hz} \leq f \leq 3.5 \text{ Hz}} HS(f). \quad (4)$$

This approach helps avoid the harmonic and subharmonic errors that may occur when detecting the heart rate using FFT spectrum. The SQI is defined as:

$$SQI = \frac{HS(f_{\text{HR}})}{\sum_{v=0.5 \text{ Hz}}^{v=20 \text{ Hz}} |\mathcal{F}_s(v)|}. \quad (5)$$

The integration of the FFT spectrum within the range of 0.5–20 Hz encompasses the energy of up to six harmonics of the possible heart rates. This SQI effectively measures the proportion of periodic components relative to the total energy within the frequency range of interest. The

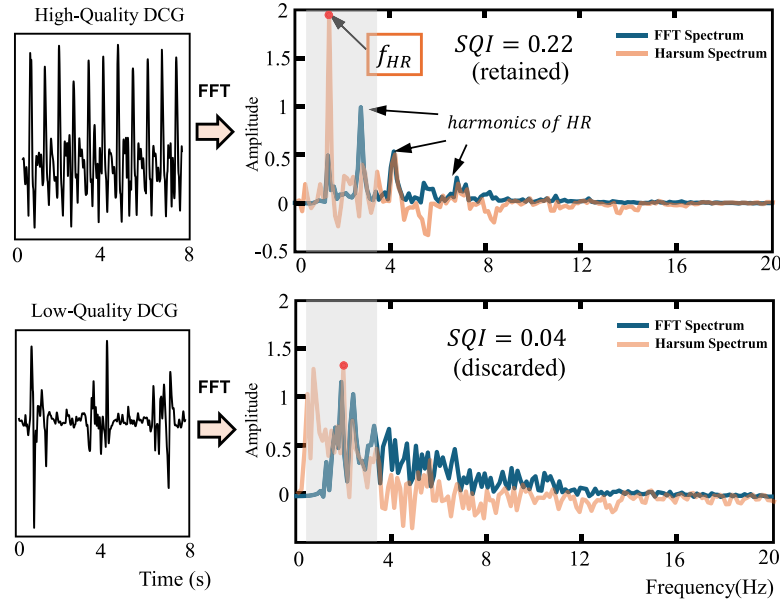


Fig. 2. Schematic diagram of HR calculation and SQI calculation.

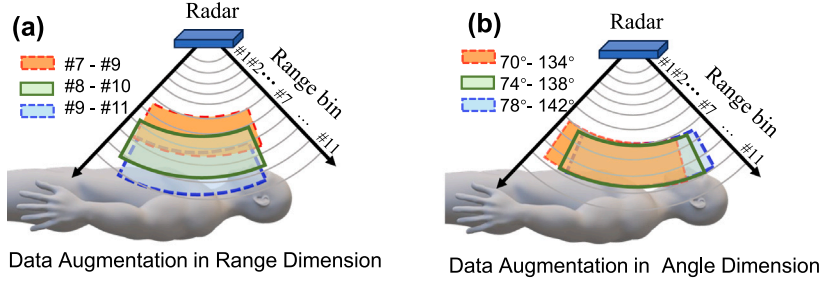


Fig. 3. Illustration of data augmentation in (a) range dimensions and (b) angle dimensions for DCG.

SQI is calculated for each segment in the DCG array $\mathcal{M}^0$. If there exists $s_{r,a} \in \mathcal{M}^0$ such that $SQI_{s_{r,a}} > \theta$, then this $\mathcal{M}^0$ is kept; otherwise, it is removed from the dataset. The threshold θ is an empirical parameter set to 0.14 in this study. The impact of θ on model performance is discussed in Section 4.5.

2.2.3. Data augmentation

To address the variations in radar-to-body distance and the differences in height and body thickness among individuals during each measurement, we perform data augmentation in both the range and angle dimensions. Specifically, we select $\mathcal{M} \in \mathbb{R}^{3 \times 64 \times L}$ from $\mathcal{M}^0 \in \mathbb{R}^{5 \times 72 \times L}$, as illustrated in Fig. 3. This process results in a 9-fold increase in usable data, thereby enhancing the diversity of the dataset.

2.3. caPTT/caPWA information focusing

Although the chest-neck DCG array contains information related to BP, extracting BP information directly from these DCGs is challenging in a limited sample size for several reasons. Firstly, the high dimensionality of the data leads to the “curse of dimensionality”, which can result in either underfitting or overfitting, depending on the complexity of the model. Secondly, the patterns within the DCG signals are highly variable, and the correlation between these patterns and BP changes is weak. Therefore, we further purify the DCG by reducing its dimensionality while selecting information that has a strong association with BP changes.

2.3.1. caPTT information focus

Extracting PTT information through the identification of fiducial points or the computation of cross-correlation between DCGs is impractical due to the morphological variability of DCGs across different anatomical locations and individual subjects. To effectively extract information related to PTTs, we construct the Multi-Location DCG Harmonic Phase Array (MLD-HPA), derived from the DCG matrix and represented as $\Phi \in \mathbb{R}^{R \times A \times N}$, where $R = 3$, $A = 64$, and $N = 4$ represent the number of ranges, angles, and harmonics, respectively. The method for calculating Φ will be detailed below.

First, the DCG at position i , denoted as $s_i \in \mathcal{M}$, is modeled as

$$s_i = \alpha_i \delta_i [2\pi f_{HR} (t + \tau_i)] + \beta_i, \quad (6)$$

where δ_i is a quasi-periodic signal with a period of the heart rate, representing the shape of the DCG for each heartbeat, which may vary over position. α_i and β_i are the scale and offset coefficients, respectively, and they also change with the position. The delay τ_i arises due to the caPTT. By employing Fourier series expansion (FSD), this quasi-periodic signal can be decomposed into a sum of sines and cosines, each representing the signal's fundamental and harmonic components. The expression for the signal in terms of complex exponentials is given by:

$$s_i \xrightarrow{\text{FSD}} \sum_{n=-\infty}^{\infty} |C_{n,i}| e^{j2\pi n f_i} e^{j\phi_{n,i}}. \quad (7)$$

The magnitude and phase shift of the signal's n th harmonic component are denoted by $|C_{n,i}|$ and $\phi_{n,i}$, respectively. It is noteworthy that in such scenarios, the changes in the form of δ_i are discernible by the varying $|C_{n,i}|$ and $\phi_{n,i}$. For a given signal $s_i(t)$, the processing steps are

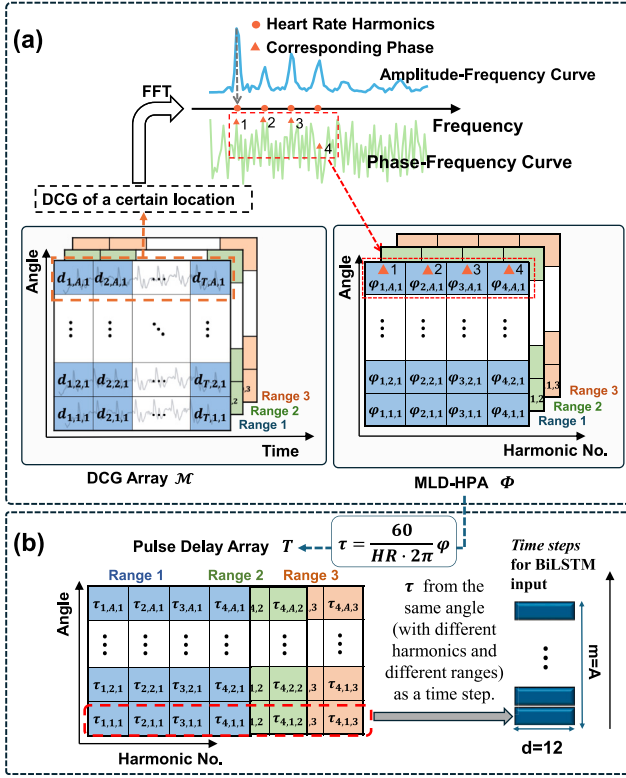


Fig. 4. Schematic diagram of (a) caPTT Information Focus and (b) Pulse Delay Array Serialization for LSTM Input.

as follows: Initially, a Hamming window is applied to $s_i(t)$. Following this, a Fourier Transform is executed to obtain the phase across various harmonic frequencies of the heart rate, denoted as

$$S_i(f) = \mathcal{F}[s_i(t)]. \quad (8)$$

The phase $\phi_{n,i}$ at the n th harmonic of the heart rate can be derived by

$$\phi_{i,n} = \arctan \left[\frac{\Im(S_i(nf_{HR}))}{\Re(S_i(nf_{HR}))} \right]. \quad (9)$$

Here, $\Im$ and $\Re$ represent the imaginary and real parts of $S_i(f)$, respectively.

As shown in Fig. 4(a), by applying the specified operations (Eq. (8) and (9)) to each $s_{r,a}$ within $\mathcal{M} \in \mathbb{R}^{R \times A \times L}$, we obtain MLD-HPA:

$$\Phi = \{\phi_{r,a,n} \mid r, a, n\}, \quad (10)$$

where $r \in \{1, \dots, R\}$, $a \in \{1, \dots, A\}$, $n \in \{1, \dots, N\}$.

Further, every $\phi_{r,a,n}$ are converted to caPTTs through the following operations related to HR:

$$\tau_{r,a,n} = \frac{60}{HR \cdot 2\pi} \phi_{r,a,n}, \quad (11)$$

where HR is obtained from the harmonic sum spectrum in Section 2.2.2.

Then we obtain Pulse Delay Array (PDA) $T \in \mathbb{R}^{R \times A \times N}$, where each element $\tau_{r,a,n}$ in T represents the phase-derived 'time delay' at the n th harmonic and position (r,a). The difference of $\tau_{r,a,n}$ between different positions can then represent caPTT.

The transformation from the DCG array to the pulse delay array is achieved as follows:

$$\mathcal{M} \in \mathbb{R}^{R \times A \times L} \rightarrow T \in \mathbb{R}^{R \times A \times N}. \quad (12)$$

2.3.2. caPWA information focus

In this section, our objective is to distill a subset of DCG channels for subsequent analysis. However, the inherent variability in pulse waveform shapes across different locations poses a challenge. Direct averaging leads to significant signal distortion, and relying on basic selection criteria, such as signal quality, may fail to capture comprehensive information. This study achieves dimensionality reduction of $\mathcal{M}$ by clustering the sequences within it.

According to the DCG model in Eq. (6), the shapes of δ_i vary according to position, but adjacent positions have similar shapes. In addition, there is a phase difference caused by PTT between these waveforms. Considering these properties, this study employs k-Shape time series clustering [24] to facilitate the dimensionality reduction of $\mathcal{M}$. K-Shape utilizes Shape-Based Distance (SBD) for distance measurement, which exhibits invariances to scale, translation, and shift and is defined as follows:

$$SBD(\vec{x}, \vec{y}) = 1 - \max_w \left(\frac{CC_w(\vec{x}, \vec{y})}{\sqrt{R_0(\vec{x}, \vec{x}) \cdot R_0(\vec{y}, \vec{y})}} \right). \quad (13)$$

Here, $\vec{x}$ and $\vec{y}$ are sequences of length m , and $CC_w(\vec{x}, \vec{y})$ represents the cross-correlation between the two-time series at an offset $s = w - m$, where $w \in \{1, 2, \dots, 2m-1\}$. The term $\sqrt{R_0(\vec{x}, \vec{x}) \cdot R_0(\vec{y}, \vec{y})}$ is a coefficient normalization factor. $SBD(\vec{x}, \vec{y})$ describes the distance between $\vec{x}$ and $\vec{y}$ after eliminating the effects of sequence shift. The centroid $\vec{\mu}_k$ of a cluster is determined by selecting the sample within the cluster that has the smallest sum of squared SBDs to all other samples within the same cluster, as formulated below:

$$\vec{\mu}_k^* = \arg \max_{\vec{\mu}_k} \sum_{\vec{x}_i \in P_k} \left(\frac{\max_w CC_w(\vec{x}_i, \vec{\mu}_k)}{\sqrt{R_0(\vec{x}_i, \vec{x}_i) \cdot R_0(\vec{\mu}_k, \vec{\mu}_k)}} \right)^2. \quad (14)$$

The steps for extracting pulse wave samples from set $\mathcal{M}$ are as follows:

Step 1: Apply the quality judgment from Section 2.2.2 to exclude the $s_{r,a}$ with an SQI less than $\theta_1 = 0.12$ from set $\mathcal{M}$. This results in:

$$\mathcal{M}' = \{s_i \in \mathcal{M} \mid SQI(s_i) \geq \theta_1\}. \quad (15)$$

Step 2: Apply k-Shape clustering to these signals to select K clusters (as described in Algorithm 1). The optimal number of clusters is determined using the silhouette coefficient method on the first 10 samples with the elbow method, and here K is set to 5.

Algorithm 1 k-Shape Based DCG Clustering

Require: DCG array $\mathcal{M}'$ composed of s_i , Number of clusters $K=5$, Max iterations $N_{max} = 150$

Ensure: Clusters $C_1, C_2, \dots, C_5$ with centroids $s_1^*, s_2^*, \dots, s_5^*$

- 1: Initialize cluster centroids $s_1^*, s_2^*, \dots, s_5^*$ randomly from $\mathcal{M}'$
- 2: **repeat**
- 3: **for** each time series $s_i \in \mathcal{M}'$ **do**
- 4: Assign s_i to the cluster C_j that minimizes the shape distance $d(s_i, s_j^*)$
- 5: **end for**
- 6: **for** each cluster C_j **do**
- 7: Update centroid s_j^* using Eq. (14)
- 8: Perform Z-normalization on s_j^*
- 9: **end for**
- 10: **until** centroids do not change or iterations reach N_{max}

Step 3: Based on the experimental setup presented in this paper, with the radar positioned directly facing the neck area, we designate the section from azimuth 85° to 95° as the neck and angles 110° to 120° as the chest. We identify the cluster with the greatest number of samples located within these respective angular regions:

$$C_{neck} = \arg \max_{C_j} N_{C_j}^{[85, 95]}, \quad (16)$$

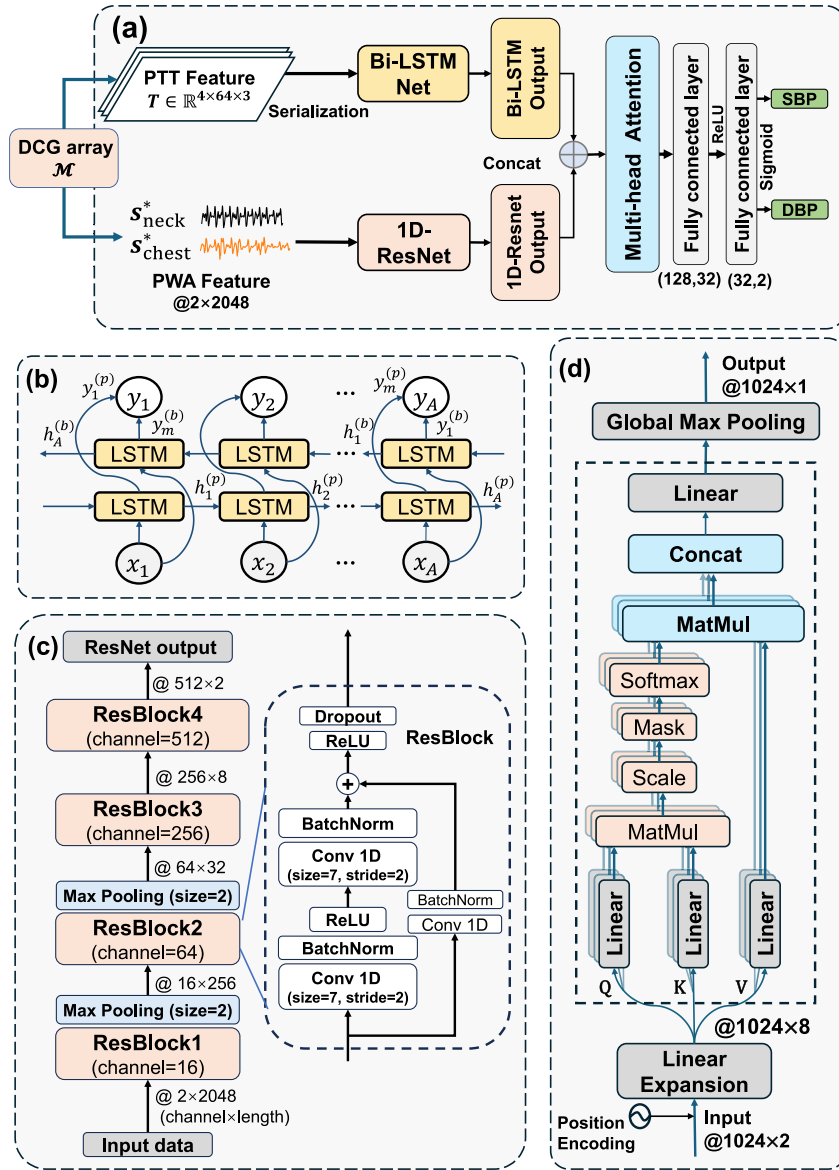


Fig. 5. Architecture of the R2S-BP Net. (a) The overall architecture of the R2S-BP Net, (b) BiLSTM structure for further extraction of caPTT features, (c) ResNet structure for extracting caPWA features, (d) Multi-head attention module for integrating the two types of features.

$$C_{\text{chest}} = \arg \max_{C_j} N_{C_j}^{[110,120]}. \quad (17)$$

Here, $N_{C_j}^{[a,b]}$ represents the number of samples in C_j that fall within a specified angle range $[a, b]$. The centroids of C_{neck} and C_{chest} , denoted as s_{neck}^* and s_{chest}^* , respectively, serve as the PWA stream inputs for the two-stream model.

2.4. BP estimation model: two-stream net

To accommodate the distinct data characteristics of PTT information and PWA information, we have developed a two-stream network architecture to extract and integrate information for BP estimation. The schematic of the proposed two-stream network (R2S-BP Net) is shown in Fig. 5(a). In this architecture, the PTT stream employs a BiLSTM structure to explore temporal-like relationships between phase differences at various locations, while the PWA stream utilizes a 1D-Convolutional Neural Network (CNN) to capture the morphological information of the pulse wave. Subsequent integration of key features is facilitated by a multi-head attention layer, followed by two fully connected layers to estimate the BP values.

2.4.1. BiLSTM-based PTT stream

Before being input into the BiLSTM, the tensor $T \in \mathbb{R}^{R \times A \times N}$ is serialized. As illustrated in Fig. 4(b), the τ from the same angle (with different harmonics and different ranges) are aggregated into a single time step. Ultimately, a time element sequence with a length of 64, where each time element has a dimension of 12, is input into a BiLSTM. Notably, this work models $\tau_{i,n}$ at different azimuth angles as different time elements in BiLSTM, as different azimuth angles actually correspond to the transit time experienced by pulse waves from the heart.

The structure of the Bi-LSTM network used is depicted in Fig. 5(b). The superscripts $\cdot^{(f)}$ and $\cdot^{(b)}$ denote the variables associated with the forward and backward phases, respectively, and h_t denotes the hidden state at the t th time step. The BiLSTM offers an advantage over the conventional LSTM by fully exploiting the bidirectional temporal features of sequences, thereby more effectively capturing information relevant to the labels (BP in this case) over both short and long time intervals [25].

As shown in Fig. 5(b), the final output at time step t can be represented as $y_t = [y_t^{(f)}; y_t^{(b)}]$, and $Y = [y_1, y_2, \dots, y_m]$, where $m = A$, indicating the number of selected angles.

2.4.2. 1D-ResNet based PWA stream

To extract waveform morphological features, we employ a 1D version of the residual network [26], as illustrated in Fig. 5(c). Initially, two 8-s segments (2000 sample points, zero-padded to 2048 points) from the DCG signal, s_{neck}^* and s_{chest}^* , are selected as 2-channel inputs for the ResNet. Subsequently, the data sequentially passes through 4 cascaded ResBlocks, each containing two 1D convolutional layers with a kernel size of 7. By setting the stride of these convolutional layers to 2 and employing pooling layers after the first two ResBlocks, we progressively reduce the dimensionality of the features. After processing through these four ResBlocks, the resulting feature map has 512 channels with a length of 2.

After the BiLSTM and 1D-CNN, two feature maps each with dimensions in $\mathbb{R}^{512 \times 2}$ are concatenated to form a single feature map with dimensions in $\mathbb{R}^{1024 \times 2}$:

$$F^{\text{concat}} \in \mathbb{R}^{1024 \times 2} = \text{Concat}(F^{\text{BiLSTM}}, F^{\text{1D-Resnet}}). \quad (18)$$

2.4.3. Multi-head attention module and output layer

We utilize a multi-head attention (MHA) module, known for its superior performance, to effectively integrate diverse features. The MHA module simultaneously attends to various aspects of the input data from multiple perspectives, enhancing both representational capacity and generalization. The structure of the constructed multi-head attention module, as shown in Fig. 5(d) can be represented as:

$$\text{Attention}(Q^i, K^i, V^i) = \text{softmax}\left(\frac{Q^i(K^i)^T}{\sqrt{d_k}}\right)V^i, \quad (19)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O, \quad (20)$$

$$\text{head}_i = \text{Attention}(QW_Q^i, KW_K^i, VW_V^i). \quad (21)$$

W_Q^i, W_K^i, W_V^i , and W^O are learnable parameters. In this work, $Q, K, V \in \mathbb{R}^{1024 \times 8}$ are obtained from the input $F^{\text{concat}} \in \mathbb{R}^{1024 \times 2}$ through a shared linear expansion layer. Position encoding is used to preserve the positional information of the original input feature maps. The final output from the multi-head attention, after passing through a global pooling layer, is $F^{\text{MHA}} \in \mathbb{R}^{1024 \times 1}$. By introducing multi-head attention, it is possible to further extract wide-term dependencies between features from PTT and PWA-related feature maps and apply higher attention weights to more important features.

The final output layer consists of a two-layer fully connected network, using a Sigmoid function to map the results to a BP value between 0 and 1. A Huber loss function is employed to reduce sensitivity to outliers [27,28]. To address potential biases in BP estimates caused by an overrepresentation of specific BP ranges in the dataset, samples are assigned different weights. The formulation of the weighted Huber loss function is presented below.

$$\mathcal{L}(y, \hat{y}) = \begin{cases} \frac{1}{2}w \cdot (y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta \\ w \cdot \delta |y - \hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise.} \end{cases} \quad (22)$$

The Huber loss function operates as follows: when the difference between the actual value y and the predicted value $\hat{y}$ is less than δ , the loss is computed using the standard mean squared error. If the difference exceeds δ , a linear absolute error loss function is employed.

Additionally, the weight w in the loss function $\mathcal{L}$ is determined based on the distribution of BP intervals within the dataset, with each 10 mmHg segment defining an interval.

$$w_i = \frac{1}{n_i^\alpha}. \quad (23)$$

Here, n_i represents the number of samples in the i th BP interval. The parameter α is a smoothing factor that controls the level of smoothing in weight distribution; $\alpha = 1$ results in weights inversely related to the number of samples, while $\alpha = 0$ gives equal weight to all samples.

Typically, α is calibrated to 0.8 via grid search to ensure an effective balance in the weighting scheme.

We utilize the same backbone network to simultaneously optimize both SBP and DBP, effectively implementing a multi-task learning approach. The aggregate loss function for this framework is defined as:

$$\mathcal{L} = \frac{1}{2}(\mathcal{L}_{\text{SBP}} + \mathcal{L}_{\text{DBP}}). \quad (24)$$

2.5. MAML-based subject-independent experiments

We conduct calibration-based subject-independent experiments through Model-Agnostic Meta-Learning (MAML) [29,30]. MAML optimizes the model by identifying a set of initial parameters that allow rapid adaptation to new tasks with few learning steps. In this work, each subject is considered as a task ($\mathcal{T}_i$) within MAML. As shown in Fig. 6, the subjects were divided into pre-training set, MAML set, and test set. For each subject in the MAML set, half of the data segments serve as the support set and the other half as the query set. The pre-training set is used to train a baseline M_ζ , and the MAML set is used to train a fine-tuned model M_ζ with optimal initial parameters $\hat{\zeta}$ that can achieve high adaptability for new subjects. After calibration with a small dataset from new subjects in the test set, the model demonstrated improved performance on the remaining data of the test set. The algorithmic process for deriving the optimal initial parameters $\hat{\zeta}$ from the initial model M_ζ using the MAML set is outlined in Algorithm 2. Overall, nine-fold cross-validation is employed to evaluate the model, with each subject in turn being part of the test set. For each subject in the test set, the model was calibrated using only four data samples from each subject, with the remaining samples reserved for testing.

Algorithm 2 MAML for calibration-based subject-independent experiments

Require: Pre-trained model M_ζ ; A batch of personalized tasks $\{\mathcal{T}_i\}$ where each task $\mathcal{T}_i$ contains all data segments of the i -th subject from MAML set, and K_i is the number of data segments of $\mathcal{T}_i$; α, β : internal and external learning rate

Ensure: Fine-tuned model M_ζ with high adaptability for new subjects

- 1: **for** all $\mathcal{T}_i \in \{\mathcal{T}_i\}$ **do**
- 2: Sample $K_i/2$ datapoints $D = \{x^j, y^j\}$ as support data.
- 3: Sample the other $K_i/2$ datapoints $D' = \{x^j, y^j\}$ as query data.
- 4: $\zeta_j \leftarrow \zeta - \alpha \nabla_{\zeta} L_i(f_{\zeta_j}, D_j)$, update the personalized parameters based on support loss
- 5: **end for**
- 6: $\hat{\zeta} \leftarrow \zeta - \beta \nabla_{\zeta} \sum_{\mathcal{T}_j} L_i(f_{\zeta_j}, D'_j)$, update the global parameters based on query losses

3. Evaluation

3.1. Dataset

The dataset used in this study is an extension of the dataset from our previous work [15]. It consists of data from 36 subjects, including 26 males and 10 females, aged between 23 and 46 years, all without a history of cardiovascular diseases. Prior to the experiment, subjects refrained from consuming alcohol and coffee. This study received approval from the Ethics Committee of the Chinese PLA General Hospital under Application No. S2022-318-05. Fig. 7 illustrates the experimental setup for data collection. A 77 GHz FMCW radar, along with its data acquisition board (AWR1642 and DCA1000EMV, Texas Instruments, USA) [31], was suspended above the subject's neck. The radar's parameter settings are provided in Table 1. A BPM-MIMO configuration, consisting of 2 transmitters (TX) and 4 receivers (RX), was utilized, resulting in a total of 8 receiving units arranged parallel to the body.

The testing protocol for each subject, shown in Fig. 7(b), involved three methods of inducing BP changes, with each session lasting approximately 30 min. Ground truth BP values were measured using a

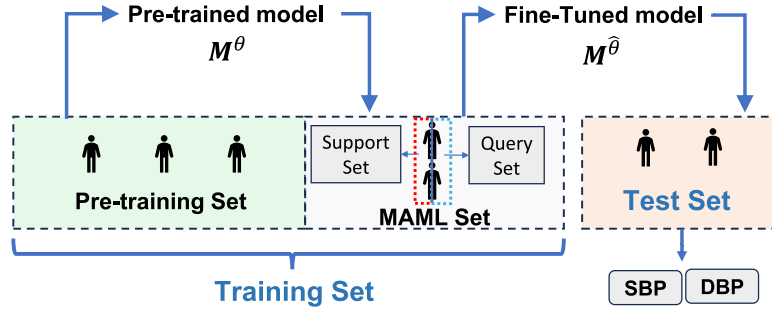


Fig. 6. Schematic diagram of the training and testing set partition.

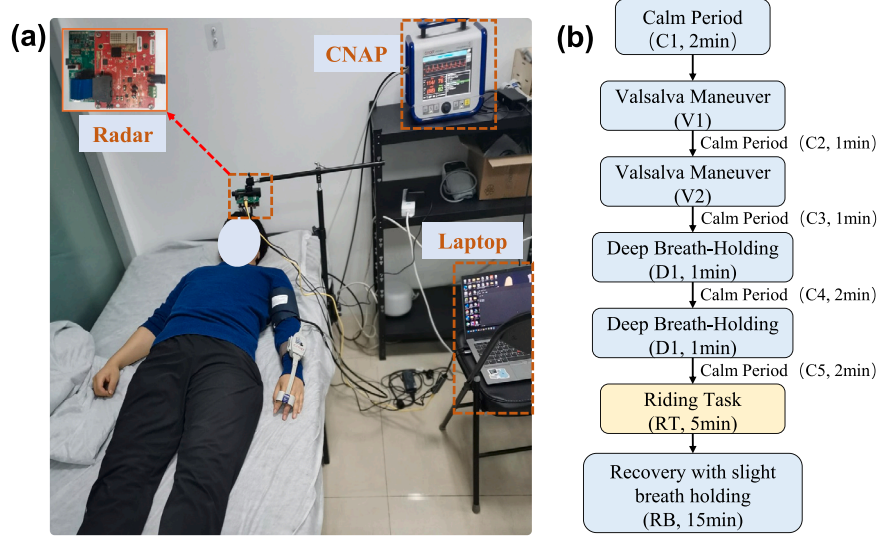


Fig. 7. (a) Test setup, (b) Testing protocol: Each subject underwent two Valsalva maneuvers, two deep breath-holding sessions, and a recovery period following a riding task. Synchronized radar data and continuous BP data were collected throughout the process, except during the riding task.

Table 1
Parameter settings of radar.

Parameter	Value
Start frequency	77 GHz
Bandwidth	4 GHz
Frequency slope	110 MHz/ μ s
Idle time	100 μ s
Ramp end time	36 μ s
Sample points	100
Sample rate	2.9 MHz
Frame periodicity	4 ms

continuous noninvasive blood pressure (CNIBP) monitor (CNAP Monitor 500, CNSystems, Graz, Austria).

The vertical distance between the radar's receiving antenna and the neck is approximately 40 cm. The radar is positioned above the neck. This placement allows for obtaining more information from the carotid artery, which is considered an optimal location for measuring pulse waves [32], as its pulsations are visibly observable to the naked eye. Additionally, compared to the carotid artery, the chest area has a larger vibrational surface area, making it easier for the radar to measure. Positioning the radar above the neck enables us to acquire superior chest vibration signals while simultaneously capturing signals from the neck.

The synchronized radar and continuous BP data were segmented into intervals of 8 s with a 2-s step size. To ensure data quality, we then excluded any segments that did not meet the required standards. This included instances where the CNAP device was in an unstable state

or radar data was flagged as low quality (see Section 2.2.2 for details). Whenever data from either the radar or CNAP did not meet the criteria, the corresponding data at that timestamp was removed. As a result, 21,707 segments were retained from an initial 29,080 segments. The distribution of SBP and DBP for these retained segments is shown in Fig. 8.

3.2. Implementation details

We implemented our deep learning architecture using PyTorch, training it on an Intel Xeon E5-2686v4 processor with 64 GB of RAM and an RTX 3090 Ti GPU with 24 GB of VRAM. The 36 subjects were divided into pre-training set, MAML set, and test set in the ratios of 18:14:4, with results presented through 9-fold cross-validation. For the test set, the model was calibrated using the first 4 data segments from each subject, with the remaining samples reserved for testing. During the pre-training phase, we used the Adam optimizer for 100 epochs with an initial learning rate of 1×10^{-3} and a decay rate of 1×10^{-4} . The dropout ratio was set to 0.2, and the batch size was 16. The turning point δ of the Huber loss is set to 0.2. In the MAML stage, both the inner learning rate α and the outer learning rate β were set to 1×10^{-3} , with a batch size of 4, trained over 4 epochs. The data augmentation techniques described in Section 2.2.3 were applied to both the pre-training and MAML sets.

3.3. Metrics

To evaluate the performance of R2S-BP, we quantitatively analyzed the results using widely accepted metrics: Mean Absolute Error (MAE),

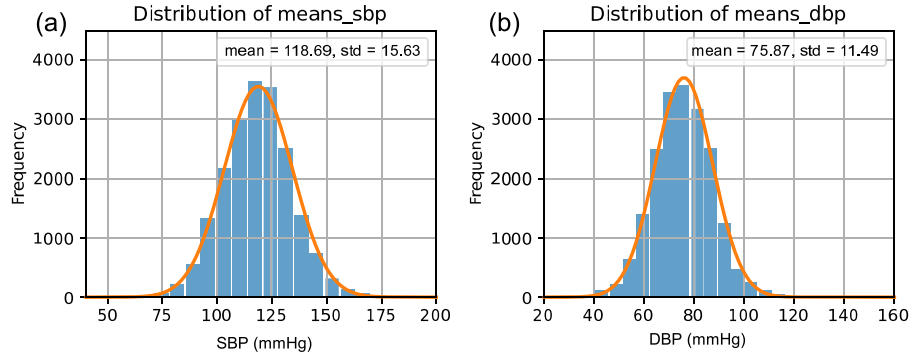


Fig. 8. Histograms of (a) SBP, (b) DBP.

Standard Deviation of Absolute Error (SD(AE)), Mean Error (ME), and Standard Deviation of Error (SD(E)). The expressions for these metrics are given as follows:

$$MAE = \frac{1}{K} \sum_{k=1}^K |y_k - \hat{y}_k|, \quad (25)$$

$$SD(AE) = \sqrt{\frac{1}{K} \sum_{k=1}^K (|y_k - \hat{y}_k| - MAE)^2}, \quad (26)$$

$$ME = \frac{1}{K} \sum_{k=1}^K (y_k - \hat{y}_k), \quad (27)$$

$$SD(E) = \sqrt{\frac{1}{K} \sum_{k=1}^K ((y_k - \hat{y}_k) - ME)^2}. \quad (28)$$

Here, y_k represents the actual value of the k -th sample, $\hat{y}_k$ denotes the predicted value for the k -th sample, and K is the total number of samples.

To evaluate the performance of BP tracking within individuals, we use the Pearson correlation coefficient to assess the correlation between the estimated values and the actual BP for each subject. The Pearson correlation coefficient r_i for each subject is calculated as follows:

$$r_i = \frac{\sum_{k=1}^{K_i} (y_k - \bar{y})(\hat{y}_k - \bar{\hat{y}})}{\sqrt{\sum_{k=1}^{K_i} (y_k - \bar{y})^2} \sqrt{\sum_{k=1}^{K_i} (\hat{y}_k - \bar{\hat{y}})^2}}, \quad (29)$$

where K_i is the number of BP data points for the i th subject. The mean of the correlation coefficients across all subjects is represented as:

$$\bar{r} = \frac{\sum r_i}{N}. \quad (30)$$

4. Results and discussion

4.1. Overall performance

Table 2 shows the overall performance compared to AAMI and BSH standards. In Table 2, CP5, CP10, and CP15 represent the Cumulative Percentages of absolute errors within 5 mmHg, 10 mmHg, and 15 mmHg, respectively.

For SBP, the results indicate a MAE of 5.54 mmHg and a SD(AE) of 4.79 mmHg, while for DBP, the MAE is 4.18 mmHg and the SD(AE) is 3.34 mmHg. As shown in Table 2, these results demonstrate that the proposed method achieves high accuracy and consistency in BP measurement, meeting the standards set by AAMI and BSH.

Table 3 presents the ground truth BP distribution for each subject alongside the corresponding estimation errors. The analysis shows that estimation errors are generally associated with the standard deviation of BP, with larger fluctuations in BP leading to greater estimation errors. Subjects 7 and 21 exhibited larger errors, which may be attributed to differences in vascular physiology and height variations, reflecting

Table 2

Overall performance of R2S-BP and comparison with BP standards.

	SBP(mmHg)	DBP(mmHg)	Standards for AAMI/BSH Grade A
MAE	5.54	4.18	–
SD(AE)	4.79	3.34	–
ME	0.04	–0.09	<5.00
SD(E)	7.33	5.36	≤8.00
CP5	69.59%	78.34%	>60%
CP10	86.56%	92.61%	>85%
CP15	95.77%	97.04%	>95%

the method's performance variability across individuals. However, the method still demonstrates effective BP tracking within subjects. For instance, even with a wider BP distribution, subjects 5 and 13 achieved low MAE. Overall, the majority of samples exhibit a SD(E) of less than 8 mmHg, and most correlation coefficients (r) exceed 0.75, with an average r of 0.82 for both SBP and DBP.

4.2. Comparison with other alternative methods

Table 4 shows the comparison of results using different methods on our dataset. The Baseline method uses the average calibration value for each subject as the estimated value. The difference between the Baseline and the Ground Truth BP shows how much the data varies, serving as a reference for other methods. The Linear Regression model is similar to our previous work, using the same calibration data but evaluating the average value over 8-s intervals instead of beat-by-beat results.

Given the obtained DCG array $\mathcal{M} \in \mathbb{R}^{R \times A \times L}$, an obvious approach is to use a 3D-CNN. In this comparison experiment, a 3D version of ResNet [33] is employed and the model input for 3D-CNN is $\mathcal{M} \in \mathbb{R}^{R \times A \times L}$. It is important to note that, given the high dimensionality of the input data (even after extracting PTT and PWA features in Section 2.3, the dimensionality remains high), and since these are sequential features rather than independent features, dimensionality reduction techniques such as PCA are not effective. As a result, traditional machine learning methods like Support Vector Regression (SVR) and Random Forest Regression (RFR) are not suitable, which is why we do not compare with them in Table 4.

Examining the SD(E) and r metrics, the Linear Regression model and the 3D-CNN model achieve similar overall results. Compared to other methods, the proposed R2S-BP demonstrates significantly superior performance across all metrics.

Fig. 9 illustrates the Bland-Altman (B-A) plots for R2S-BP and 3D-CNN. In the B-A plots, the red dashed lines represent the 95% confidence intervals of the errors, which is the mean value ± 1.96 SD. The black dashed line indicates the mean error. For clarity, the horizontal axis represents the ground truth BP, and different colors are used to

Table 3

Distribution of BP for each subject and BP estimation results of R2S-BP.

ID	SBP						DBP					
	Mean $\pm$ STD	MAE	SD(AE)	ME	SD(E)	<i>r</i>	Mean $\pm$ SD	MAE	SD(AE)	ME	SD(E)	<i>r</i>
1	124.78 $\pm$ 12.68	5.54	4.78	1.07	7.24	0.81	88.26 $\pm$ 8.46	3.94	2.81	2.36	4.22	0.84
2	104.05 $\pm$ 9.41	4.57	3.60	-2.17	5.40	0.81	69.49 $\pm$ 7.75	3.57	2.79	-1.87	4.13	0.82
3	118.99 $\pm$ 12.21	5.68	4.38	-1.98	6.89	0.83	72.15 $\pm$ 14.09	4.27	3.33	-2.10	4.99	0.89
4	105.85 $\pm$ 11.98	4.53	3.79	-1.35	5.75	0.87	70.34 $\pm$ 10.28	3.81	2.98	-1.53	4.59	0.86
5	132.52 $\pm$ 15.54	6.00	4.74	3.28	6.90	0.86	83.91 $\pm$ 8.89	4.49	3.42	-0.40	5.63	0.79
6	129.71 $\pm$ 13.10	5.35	4.45	2.03	6.66	0.84	85.12 $\pm$ 7.95	3.57	2.60	1.78	4.04	0.83
7	116.72 $\pm$ 12.21	11.07	9.19	-1.97	14.25	0.57	78.84 $\pm$ 8.02	8.19	5.84	0.07	10.06	0.53
8	125.10 $\pm$ 12.08	5.63	4.65	1.03	7.23	0.82	85.10 $\pm$ 8.66	3.71	2.78	2.05	4.16	0.84
9	133.53 $\pm$ 16.61	5.70	4.90	2.40	7.12	0.85	79.04 $\pm$ 9.67	4.30	3.04	0.61	5.23	0.82
10	126.81 $\pm$ 14.24	6.00	5.10	2.39	7.50	0.81	86.40 $\pm$ 8.44	4.84	3.61	-0.79	5.99	0.75
11	123.42 $\pm$ 8.34	4.42	3.67	0.37	5.73	0.78	85.08 $\pm$ 5.41	3.72	2.67	2.62	3.76	0.79
12	115.61 $\pm$ 10.76	4.93	3.96	-1.85	6.04	0.83	75.63 $\pm$ 7.87	3.43	2.57	-0.48	4.26	0.82
13	125.20 $\pm$ 15.71	5.76	4.66	0.74	7.37	0.84	76.82 $\pm$ 12.17	4.31	3.26	-0.20	5.40	0.87
14	119.64 $\pm$ 13.38	5.45	4.43	-0.91	6.96	0.83	77.51 $\pm$ 10.86	3.86	2.89	0.00	4.82	0.86
15	121.57 $\pm$ 11.51	5.64	4.55	0.97	7.19	0.78	76.03 $\pm$ 10.78	4.84	3.67	-0.02	6.07	0.82
16	116.05 $\pm$ 13.76	5.46	4.46	-1.76	6.83	0.83	71.78 $\pm$ 10.21	4.36	3.31	-1.55	5.25	0.85
17	132.31 $\pm$ 18.83	6.82	5.86	3.15	8.42	0.86	81.22 $\pm$ 13.14	4.23	3.11	1.27	5.10	0.88
18	101.58 $\pm$ 9.90	4.40	3.60	-1.81	5.39	0.83	72.44 $\pm$ 9.64	3.68	2.76	-1.27	4.42	0.85
19	131.33 $\pm$ 16.00	6.14	5.02	2.67	7.47	0.85	88.62 $\pm$ 8.74	3.97	2.85	2.44	4.23	0.84
20	113.16 $\pm$ 11.01	5.76	5.02	-0.58	7.62	0.76	74.09 $\pm$ 13.90	4.34	3.59	0.39	5.62	0.87
21	122.82 $\pm$ 16.29	10.07	8.29	2.83	12.73	0.71	75.03 $\pm$ 9.19	8.10	5.83	0.38	9.97	0.65
22	123.22 $\pm$ 13.71	5.08	4.01	0.47	6.45	0.85	78.99 $\pm$ 11.84	3.82	3.12	0.34	4.92	0.86
23	117.59 $\pm$ 13.96	5.16	3.98	-2.07	6.17	0.85	73.52 $\pm$ 12.39	3.92	2.90	-1.21	4.72	0.88
24	137.28 $\pm$ 15.26	6.20	4.80	2.91	7.28	0.84	80.11 $\pm$ 9.28	3.45	2.55	0.76	4.22	0.87
25	131.25 $\pm$ 12.27	6.02	4.86	4.38	6.38	0.83	83.73 $\pm$ 11.68	4.83	3.73	-0.15	6.10	0.82
26	128.38 $\pm$ 15.37	6.27	4.87	2.14	7.65	0.84	77.37 $\pm$ 8.64	3.47	2.68	-0.08	4.39	0.83
28	108.92 $\pm$ 9.78	4.41	3.57	-1.72	5.40	0.84	63.96 $\pm$ 9.74	3.65	2.86	-2.16	4.10	0.87
29	115.25 $\pm$ 12.16	5.31	4.47	-0.81	6.90	0.79	74.10 $\pm$ 7.28	4.02	2.88	-0.42	4.92	0.78
30	107.06 $\pm$ 11.15	5.73	4.65	-1.60	7.21	0.75	79.76 $\pm$ 7.64	4.47	3.44	0.09	5.64	0.72
31	122.94 $\pm$ 12.93	5.27	4.21	0.23	6.74	0.82	76.46 $\pm$ 6.48	3.78	2.83	0.01	4.72	0.76
32	113.80 $\pm$ 10.29	4.55	3.56	-2.11	5.37	0.84	76.46 $\pm$ 9.04	3.64	2.82	0.31	4.59	0.85
33	121.80 $\pm$ 13.37	5.18	4.35	-0.38	6.75	0.84	81.43 $\pm$ 9.75	3.75	2.86	0.97	4.61	0.84
34	111.50 $\pm$ 6.51	4.47	3.41	-1.56	5.40	0.84	74.48 $\pm$ 7.57	3.70	2.77	-0.15	4.62	0.81
35	117.50 $\pm$ 16.84	5.73	4.45	-1.56	7.08	0.83	70.67 $\pm$ 13.92	4.43	3.43	-0.59	5.57	0.87
36	112.92 $\pm$ 8.39	3.94	3.05	-2.43	4.34	0.80	68.12 $\pm$ 10.93	4.04	2.94	-2.10	4.53	0.86
Mean	119.77 $\pm$ 12.46	5.59	4.61	-0.16	7.17	0.82	77.09 $\pm$ 10.08	4.10	3.04	-0.32	4.72	0.82

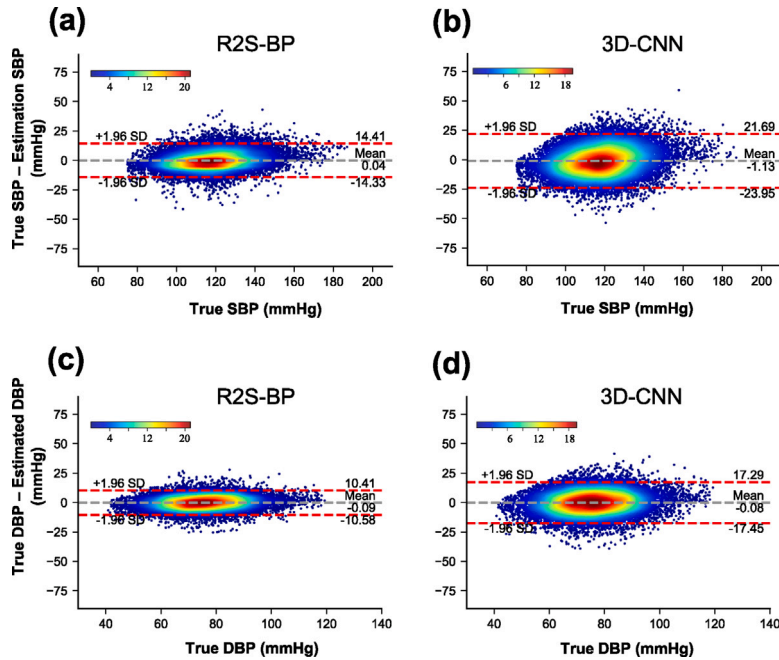
**Fig. 9.** Bland-Altman plots of (a) R2S-BP estimated SBP, (b) 3D-CNN estimated SBP, (c) R2S-BP estimated DBP, and (d) 3D-CNN estimated DBP.

Table 4

Performance comparison of different methods in the same dataset.

Model	SBP					DBP				
	MAE	SD(AE)	ME	SD(E)	$\bar{r}$	MAE	SD(AE)	ME	SD(E)	$\bar{r}$
Baseline (Calibration value)	13.80	11.22	4.32	17.25	0.24	9.70	8.10	-2.72	12.21	0.31
Linear regression	9.58	7.45	3.96	11.43	0.72	7.91	6.86	2.93	9.73	0.69
3D-CNN	9.28	7.12	-1.13	11.64	0.70	6.95	5.50	-0.08	8.86	0.71
R2S-BP (Proposed)	5.54	4.79	0.04	7.33	0.82	4.18	3.34	0.09	5.36	0.82

Table 5

Results of ablation study.

Model	SBP					DBP				
	MAE	SD(AE)	ME	SD(E)	$\bar{r}$	MAE	SD(AE)	ME	SD(E)	$\bar{r}$
Complete model	5.54	4.79	0.04	7.33	0.82	4.18	3.34	-0.09	5.36	0.82
w/o PTT stream	9.98	8.11	1.26	12.20	0.65	7.89	6.82	-0.93	9.98	0.66
w/o PWA stream	7.46	7.02	2.28	10.43	0.76	7.65	7.43	-1.43	10.57	0.59
w/o k-Shape	7.71	7.89	-0.86	11.85	0.69	9.16	7.80	2.26	11.06	0.48
w/o MAML	9.98	8.19	1.47	12.45	0.49	8.02	7.60	1.56	10.56	0.38
w/o few-shot calibration	11.89	10.05	-2.86	12.70	0.53	7.65	6.05	2.26	9.06	0.51

show the density of sample points. It can be observed that the R2S-BP generally has smaller errors compared to 3D-CNN, demonstrating the effectiveness of the proposed method, particularly the physiologically guided manual extraction of caPTT and caPWA features. Additionally, in the lower BP range, the estimated BP values are higher than the true BP values, while in the higher BP range, the estimated values are lower than the actual BP values. This occurs because deep learning methods tend to output middle values, despite applying the weighted loss.

4.3. Ablation study

To evaluate the effectiveness of key components in R2S-BP, we conducted ablation experiments. In Table 5, we present the results of the complete R2S-BP model, as well as models without the PTT stream (w/o PTT stream), without the PWA stream (w/o PWA stream), and without the k-Shape component (w/o k-Shape) for both SBP and DBP. In the w/o PTT stream and w/o PWA stream settings, the PTT stream and PWA stream were removed from the model, and the input dimension parameters for subsequent fusion and output layers were adjusted accordingly. In the w/o k-Shape setting, the input for the PWA stream was selected as the signal with the highest SQI from the DCG array $\mathcal{M}$ in the chest region (angles 85–95) and neck region (angles 110–120). All settings were evaluated in the independent subject scenario under the MAML setting (except for the w/o MAML setting).

As shown in Table 5, removing the PTT stream module significantly increased the evaluation error. Compared to the complete model, the SD(E) for SBP increased from 7.33 to 12.20 mmHg, and for DBP, it increased from 5.36 to 9.98 mmHg. This highlights the importance of the extracted PTT features. Fig. 10 shows the SBP and DBP variations for individual subjects under different conditions. The complete R2S-BP model tracks BP changes well, while the model without the PTT stream struggles with sudden changes and exhibits greater variance.

In the w/o PWA Flow setting, the SD(E) for SBP increased from 7.33 to 12.20 mmHg, and for DBP, it increased from 5.36 to 9.98 mmHg. Comparing the results of the w/o PTT Flow and w/o PWA Flow settings, the SBP performance drops more significantly without the PTT Flow, indicating a greater increase in SD(E). Conversely, the DBP performance drops more significantly without the PWA Flow, suggesting that the PTT Flow is more critical for SBP estimation, while the PWA Flow is more critical for DBP estimation. Nevertheless, the PTT stream contributes the most to the overall performance of the model. In our previous work [15], a strong correlation was observed between radar-measured caPTT and contact caPTT, both of which reliably tracked BP fluctuations. This result further elucidates and highlights the critical role of the PTT stream in the current study.

The w/o k-Shape setting shows even worse results than the w/o PWA Flow setting (SD(E)=11.85 mmHg) because the unselected PWA waveforms introduced greater uncertainty into the model.

4.4. The role of MAML training strategy

This section primarily examines the role of the MAML training framework in our study. While a typical approach involves using the Leave-One-Subject-Out (LOSO) method for training and evaluating models, our experiments revealed challenges with LOSO when integrating calibration data. Specifically, the calibrated model tended to exhibit biases toward or close to the calibration values.

As shown in Table 5, the performance of the model without MAML integration into the training set, particularly in terms of the correlation coefficient $\bar{r}$, significantly declined. This highlights the critical role of MAML in refining initial parameter settings and ensuring model generalizability.

The introduction of MAML addresses this challenge by maintaining separation between training and test data while optimizing initial parameters to better adapt to new tasks without overfitting to prior data. Research has demonstrated that MAML acts as an implicit supervised contrastive learner, leveraging support features as positive and negative samples to guide model updates [34]. This characteristic helps the model maintain balance and enhances its ability to generalize, particularly in real-world scenarios where new data may vary substantially from the training set.

As shown in Table 5, the removal of few-shot calibration (w/o few-shot calibration) also resulted in performance degradation. Compared to the w/o MAML condition, the decline in performance without few-shot calibration was evident in numerical errors for BP while showing better performance in terms of the correlation coefficient $\bar{r}$. This suggests that while few-shot calibration has a significant impact on error metrics, its primary function is to approximate the blood pressure range for a given subject.

4.5. Influence of SQI threshold (θ)

The SQI threshold (θ), introduced in Section 2.2.2, determines the quality level of data used for model training and evaluation. Given that DCG signals frequently contain low-quality data, the selection of θ is critical. In this section, we conducted additional experiments to assess the impact of θ on model performance. Specifically, we selected three different θ values to filter the training data, resulting in three distinct models. These models were then evaluated on test datasets filtered using six different θ values. The resulting SD(E) values for SBP and DBP are shown in Fig. 11(a) and (b). The proportion of data retained after filtering with varying thresholds relative to the original dataset is presented in Fig. 11(c).

The results reveal that higher SQI thresholds in the test dataset generally lead to better performance, as evidenced by lower SD(E)

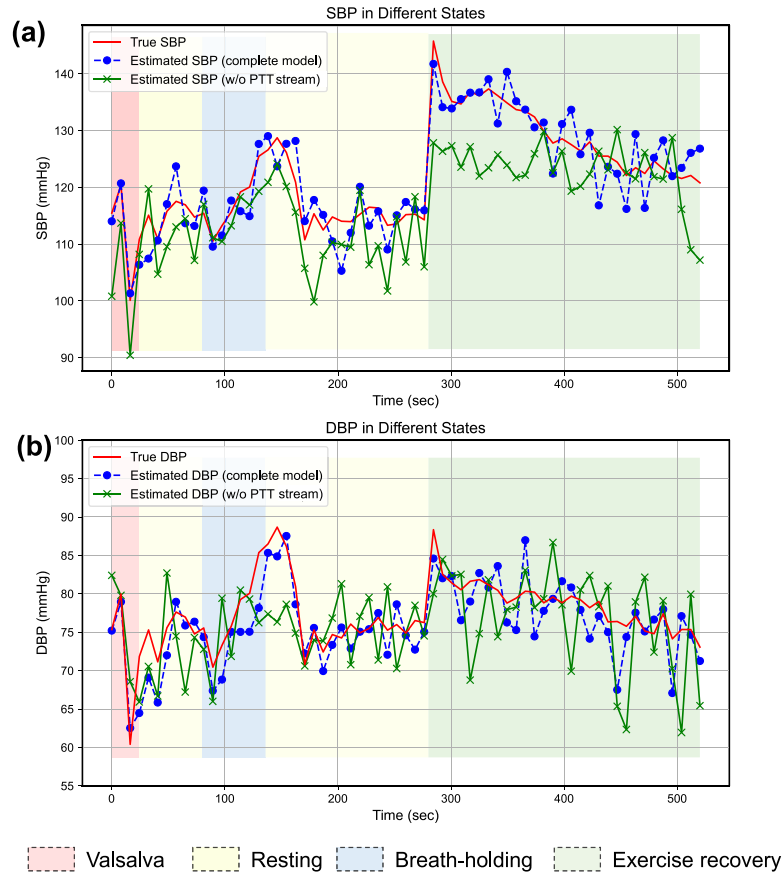


Fig. 10. Comparison of BP Tracking Capability between the Complete Model and the Model without the PTT Stream. (a) SBP. (b) DBP.

Table 6
Comparison with other works.

Year	Work/Author	Setting					Method		Result(mmHg)	
		Radar	Position	Subject	Induction	Subject independent	Input	Estimator	SBP	DBP
2022	mmBP [13]	FMCW	Wrist	25	No	Yes	WF	RF	SD=5.01	SD=5.27
2023	hBP-Fi [35]	FMCW	Arm	35	No	No	Waveform	CNN	SD=6.83	SD=6.30
2023	airBP [36]	FMCW	Wrist	41	1	Yes	Waveform	CNN	SD=4.80	SD=3.79
2022	Zheng [37]	CW×2	Chest&Wrist	27	1	Yes	PTT&WF	MLP	SD=12.13	SD=8.12
2023	TRCCBP [38]	UWB	Chest	31	2	Yes	Waveform	CNN	MAE=4.49	MAE=4.73
2024	Qiu [14]	CW	Chest	30	3	No	Waveform	DCN	SD=6.14	SD=5.50
2024	Vysotskaya [39]	FMCW	Chest	55	No	No	WF	Transformer	SD=1.77	SD=2.36
2024	R2S-BP (Ours)	FMCW	Chest&neck	36	3	Yes	PDA&waveform	2stream net	SD=7.33	SD=5.36

WF:Waveform Features; CNN:Convolutional Neural Network; MLP:Multilayer Perceptron; DCN:Deformable Convolution Network; PDA: Pulse Delay Array.

values. Signals with higher SQI values typically contain fewer artifacts and noise, enabling the model to capture the subtle features necessary for accurate BP estimation. However, when the test dataset's threshold exceeds a certain value, the performance reaches a plateau, indicating that signal quality may no longer be the primary factor affecting the model's performance.

On the other hand, interesting patterns emerge when varying the SQI threshold for the training dataset. Models trained with a specific threshold ($\theta = 0.14$) achieve the best performances. This observation suggests that excessive noise in the training data can negatively affect the model's performance by introducing bias, while overly strict filtering of low-quality signals can reduce the diversity of the training dataset, potentially impairing the model's ability to generalize to more diverse real-world data.

These findings highlight the existence of an optimal SQI threshold that balances signal quality and dataset diversity to maximize model performance. Future work could explore adaptive SQI strategies to further optimize performance across varying conditions.

4.6. Comparison with other work

Table 6 presents the details and results of several representative works using radar for non-contact BP measurement. It is important to note that the purpose of this comparison is to provide a comprehensive reference for research in this field. Directly comparing results across different studies can be challenging due to several influencing factors:

Firstly, there are differences in the methods used to induce BP changes. When BP change induction is insufficient or not performed, individual BP changes are minimal, leading to small errors in estimators. This does not fully reflect the estimators's effectiveness. According to Mukkamala's recommendation [40], claims of meeting AAMI or BHS standards should only be made with adequate BP change induction, involving at least three different induction methods. Furthermore, data leakage can significantly affect results. When independent subjects are not used, and the training and testing sets are randomly divided, models may learn specific features related to individual subjects rather than features associated with BP. Note that, although our work did not

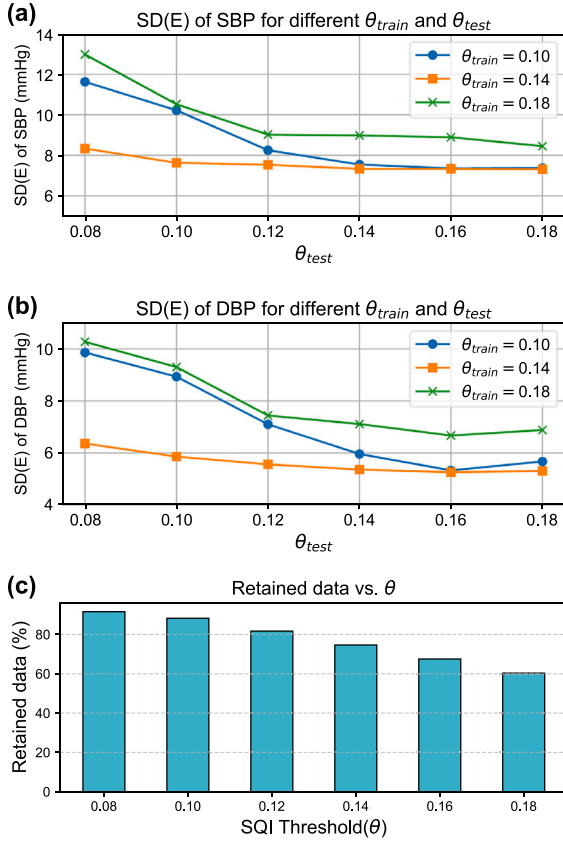


Fig. 11. Model performance under different SQI thresholds. (a) SD(E) of SBP for various θ_{train} and θ_{test} . (b) SD(E) of DBP for various θ_{train} and θ_{test} . (c) Retained data proportion as a function of θ .

achieve the best results, it is currently the only study that meets the AAMI standard's error requirements with independent subject experiments under an experimental setup involving three types of BP change induction. However, it is important to note that our study does not meet the required sample size or use the recommended reference BP measurements (a common limitation in the field).

Additionally, the increasing prevalence of research utilizing FMCW radar can be observed in Table 6, primarily due to its distinct advantages over CW radar. FMCW radar offers range resolution and can utilize MIMO antennas to achieve angle resolution, allowing for the acquisition of high SNR heartbeat signals from selected positions at greater distances. More importantly, but often overlooked in other studies, is FMCW radar's capability to capture heartbeat signals from multiple locations. In Table 6, studies using CW and UWB radar are limited to obtaining single-channel DCGs, while most FMCW radar studies typically analyze DCG signals from a single location to estimate BP using PWA methods. Ref. [37] is an exception, employing PTT and waveform features as inputs but utilizing two CW radars to achieve this. In contrast, our work employs a single FMCW radar to measure pulse waves from multiple locations, thereby enhancing the robustness and accuracy of BP estimation without adding extra hardware burden.

In conclusion, we believe that beyond the numerical results, the superior hardware and algorithm design, as well as the more rational experimental design and evaluation methods employed in this paper, also provide valuable insights and inspiration for future research in this field.

4.7. Potential advantages and applicability

Unlike wrist-based measurement setups, this study uses FMCW radar to measure the central artery in a supine position, which holds

promise for monitoring BP during sleep. Although this study only measured BP in specific postures, previous research has demonstrated that similar radar parameter configurations can achieve sleep posture recognition [41]. Consequently, BP monitoring could be conducted when specific postures are detected.

Additionally, the central artery, with its minimal smooth muscle content, is less influenced by nervous system regulation. This suggests that, for the same subject, additional calibration may not be required for up to six months [7].

In terms of electromagnetic safety, the total maximum transmission power of the antenna is only 12 dBm, which is approximately 1/60th of the Federal Communications Commission's (FCC) maximum allowed transmission power for WiFi, set at 30 dBm. This is also well below the International Commission on Non-Ionizing Radiation Protection (ICNIRP) guidelines for RF electromagnetic field exposure in the 6–300 GHz range [42]. This indicates that FMCW radar for BP monitoring can be very safe.

4.8. Limitations

According to the AAMI and AAMI/ESH/ISO universal standards, experiments should include at least 85 subjects [43]. However, the sample size in this study is considerably smaller than the recommended number. While variations in BP were introduced through induced changes, the limited sample size may restrict a comprehensive evaluation of the method's performance. Furthermore, the AAMI/ESH/ISO universal standard requires reference BP measurements to be made using a mercury sphygmomanometer or a highly accurate non-mercury sphygmomanometer. To better validate the method, future studies should consider increasing the sample size and using the recommended reference BP measurements.

The sample in this study included only healthy individuals and lacked samples from those with pathological hypertension or hypotension. These populations may have different physiological conditions compared to healthy subjects, raising questions about the generalizability of the results. However, the ability to track BP in healthy individuals has been validated, which is useful for analyzing nocturnal BP trends and variability. Future studies should include more patient samples.

The experiments in this study were conducted with the radar positioned relatively close to the body (40 cm), limited by the radar's angular resolution. In the future, using radars with a larger number of antennas could achieve higher angular resolution, allowing for the distinction of DCG signal origins from different body parts at greater distances.

Although the computational load has been significantly reduced compared to the 3D-CNN approach, future work should explore model compression techniques to further reduce hardware resource requirements, enabling deployment on portable devices.

5. Conclusions

Estimating BP using central arterial and cardiac activity information obtained from FMCW radar is a promising method. Building on our previous validation work [15], this paper proposes a two-stream framework for BP estimation, designed to analyze multi-site DCG data measured by a single FMCW radar. By carefully crafting input features and employing a two-stream network estimator to handle caPTT and caPWA information, effective BP tracking is achieved. Extensive experiments on a dataset collected using three different BP induction methods demonstrate that the proposed method achieves superior estimation performance, meeting the AAMI standard and attaining grade A of the BHS standard. And it significantly outperforms other available models.

In conclusion, this work introduces a novel tool for non-contact BP estimation based on FMCW radar. Future research will focus on long-term measurements, collecting samples from hypertensive patients, and using larger antenna arrays to achieve measurements at greater distances, further advancing the practical application of radar-based unobtrusive BP monitoring.

CRediT authorship contribution statement

Zhongrui Bai: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fanglin Geng:** Validation, Data curation. **Hao Zhang:** Writing – review & editing, Conceptualization. **Xianxiang Chen:** Writing – review & editing, Validation. **Lidong Du:** Writing – review & editing. **Peng Wang:** Project administration. **Pang Wu:** Writing – review & editing, Software, Resources. **Gang Cheng:** Resources. **Zhen Fang:** Writing – review & editing, Supervision, Funding acquisition. **Yirong Wu:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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